

Platoon Control Leveraging Network Performance and State Estimation Under Dynamic V2V Network

Jun Gao¹, Zhiyuan Peng, Soohyun Jang², and Changhao Piao¹

Abstract—This paper presents a distributed platoon control method designed to mitigate the impact of dynamic network interruptions on vehicle platoon safety and stability. Such interruptions pose significant challenges to maintaining platoon formation and achieving effective tracking performance. The proposed approach integrates network performance and state estimation within a nonlinear distributed model predictive control (DMPC) framework. A V2V communication model for platooning on straight roads is developed using stochastic geometry theory, and a unidirectional connectivity determination method assesses platoon topology connectivity. Missing information due to communication degradation is estimated using the probabilities of successful V2X information reception and connectivity results. This estimation provides reference trajectories for disconnected vehicles, ensuring platoon safety and stability. The performance of the proposed method is evaluated through simulations of a heterogeneous platoon with 15 vehicles under nine network topologies, considering both fixed-time and random intermittent network interruptions. Results demonstrate that under random network interruptions, the proposed method improves tracking performance by 87.1%, driving comfort by 72.5%, and energy efficiency by 23.2% compared to a baseline control method. Preliminary real-world testing further validates the feasibility of the approach. These findings underscore the effectiveness of the proposed method in maintaining platoon stability and safety, even during complete V2X communication loss.

Index Terms—Vehicle platooning, V2V network performance, network connectivity, state estimation, predictive control.

I. INTRODUCTION

VEHICLE platooning has attracted increasing attention due to its potential to improve traffic efficiency and

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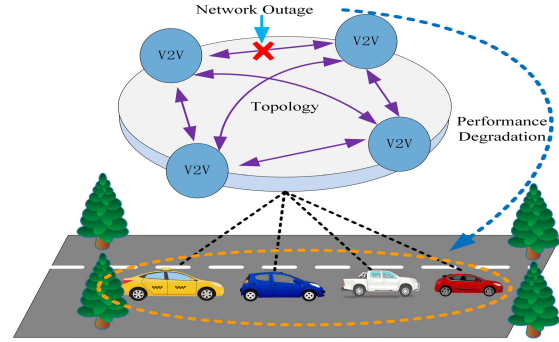


Fig. 1. Cooperative scenarios of platooning under V2V communication failure.

reduce energy consumption through close-gap coordination. The main goal of platoon control is to synchronize vehicle speeds and maintain short inter-vehicle distances. Connected vehicle platooning technology is crucial for intelligent vehicles and transportation systems. Utilizing V2X (vehicle to everything) communication, it enhances data sharing and perception capabilities [1], [2]. Common control approaches include classical proportional-integral-derivative (PID) control [3], model predictive control (MPC) [4], linear-quadratic control [5], H_∞ control [6], sliding mode control [7], and various intelligent control techniques [8]. However, imperfect communication networks compromise platoon stability, comfort, and energy efficiency. As a result, considerable research has been devoted to ensuring stable platoon behavior under non-ideal network conditions. Fig. 1 depicts a scenario where V2V (vehicle to vehicle) communication failure degrades platoon performance. Issues such as latency, packet loss, or disconnection can undermine the platoon's safety and stability. This highlights the need to develop control strategies that ensure stability under imperfect network conditions.

Communication delay is one of the most common and influential factors affecting the stability of vehicle platoons [9]. Existing compensation methods for V2V delays generally fall into two categories: physics-based and data-driven approaches [10]. Kalman filtering (KF), a typical model-based technique, has been widely used for compensating communication delays. For instance, Tan and Zhang [11] built a KF-based vehicle state prediction model that fuses sensor and wireless data to estimate delays. Wang et al. [12] applied an extended Kalman filter to address delay and stability issues. Liu et al. [13] introduced a Smith predictor-based (SP) CACC (cooperative adaptive cruise control) system to mitigate delay effects and enhance platoon robustness.

While physics-based methods are easy to implement, their performance depends on model accuracy. To address model limitations, learning-based methods have emerged [14], [15], [16]. In [17] and [18], an LSTM (long short-term memory) network was used to predict future acceleration, which was then shared for delay compensation. Zhang et al. [19] combined Bi-LSTM with MPC to maintain stability even under severe delay conditions. Hsueh et al. [20] proposed a hybrid model based on LSTM and high-order polynomial regression with an attention mechanism for feature extraction. Despite strong predictive capability, neural network-based methods rely heavily on data quality and quantity, and are prone to overfitting and limited real-world adaptability. Some studies have also explored event-triggered V2V communication to optimize network performance [21], [22]. For example, [23] used local and neighboring vehicle data to trigger transmissions, reducing the need for continuous monitoring and saving bandwidth. However, such work often lacks a clear strategy for maintaining stability once delays occur. Moreover, to simplify modeling, some approaches assume known delays or use heuristics like timestamp thresholds for delay detection. These simplifications can introduce compensation errors and degrade platoon performance.

In addition to delay, communication link failures and dynamic topology changes have garnered attention [24]. Several studies aim to maintain platoon stability by adjusting information acquisition mechanisms. For example, Markov chains have been used to model topology switching due to vehicle mobility or network degradation [25]. Weighted correction strategies [26] and multi-objective optimization approaches [27] have been proposed to enhance data accuracy and control performance. The concept of average dwell time has also been introduced to guarantee internal system stability under switching topologies [28].

While extensive efforts have been made to address platoon control under communication delays and dynamic topology changes, communication interruptions, common in real-world vehicular networks, have received comparatively less attention. Existing studies typically resort to adaptive cruise control (ACC) fallback strategies to ensure safety during connectivity loss [29]. Although such mechanisms can prevent collisions, they are often inadequate in maintaining platoon stability and desired inter-vehicle spacing. This gap highlights the necessity of developing robust control frameworks and cooperative data compensation schemes specifically designed for communication interruption scenarios.

Current research primarily focuses on evaluating communication-layer network performance, often neglecting the impact of V2X performance (e.g., success probability of information transfer) on platoon metrics such as stability, energy efficiency, and comfort. Additionally, a systematic method to assess V2X performance within platoons is lacking. While studies on message acquisition focus on network topology, they overlook the random changes in information links caused by vehicular mobility and self-organization. Therefore, we propose a vehicle platoon control method that considers V2V network performance and collaborative data compensation.

This study distinguishes itself from previous research in several key aspects.

- i) This study innovatively combines the evaluation of platoon communication performance, state estimation of the leader vehicle, and distributed model predictive control (DMPC) within a dynamically changing network environment. A model is established to estimate the success probability of V2V network communication within the platoon, thereby assessing the likelihood of vehicles successfully receiving information.
- ii) A unidirectional network topology connectivity determination method is introduced into the platoon control model to assess the connectivity of the current platoon network. The probability of successful information reception and network connectivity results are then used to estimate collaborative data, compensating for the loss of V2V shared information, ensuring both vehicle and platoon stability and safety.
- iii) Four performance indicators are employed to quantitatively evaluate the effectiveness of the proposed control method, assessing the performance of the platoon under fixed-time and random network interruption scenarios. This comprehensive approach addresses the challenge of maintaining platoon stability during random network interruptions or intermittent V2V network losses.

The structure of this paper is as follows: Section II presents models for evaluating V2V network performance and connectivity in platooning. Section III develops models for platooning and collaborative state estimation. Section IV designs a DMPC controller and analyzes platoon stability. Section V presents simulations, and Section VI provides experimental validation. Finally, Section VII concludes the study.

Additionally, some frequently used symbols are listed in Table I.

II. MODELING OF V2V COMMUNICATION

Connected vehicles utilize V2V communication to exchange information on the road. With a sufficiently large coverage area and a significant number of vehicles, a macroscopic V2V communication model can be developed using these transmitting and receiving nodes. Preliminary mathematical assumptions, based on stochastic geometry theory, will be made for V2X communication in vehicle platoons using Poisson line processes (PLP) and Poisson point processes (PPP). The analysis will focus on the success transmission rate, considering PLP/PPP, line density, and point density.

Definition 1 ([30]): (PLP). As shown in Fig. 2, an undirected line $L_i \in \mathbb{R}^2$ is defined by the perpendicular distance ρ_i from the origin to the line and the angle θ_i between the perpendicular extension and the positive x-axis. Each line in $\mathbb{R}^2$ can be uniquely mapped to the cylindrical plane $C \equiv \mathbb{R}^+ \times [0, 2\pi)$ with coordinates (ρ_i, θ_i) . The set of random lines generated according to the PPP is called the PLP.

Definition 2 ([30]): (Line density). Line density in linear processing is defined as the average length of line segments per unit area.

TABLE I
SOME SYMBOLS USED IN THIS STUDY

Symbol	Description	Symbol	Description
$\ \cdot\ $	Euclidean norm	P_r	V2V receive power
$L_i \in \mathbb{R}^2$	Undirected line	ψ_l	Poisson line process
ψ_C	Poisson point process	P_t	V2V transmitting power
G_t, g_t	Main or side lobe gains	$\mathbb{P}_{succ}$	Probability of successful reception
p_i^t	Vehicle position	$v_{est,t}^t$	Velocity
$\Delta_t = 0.1s$	Sample time	$m_{veh,i}$	Vehicle mass
$\eta_{veh,i}$	Driveline efficiency	$R_{rad,i}$	Tire radius
$\mathbf{P}_t$	System error covariance matrix	$\mathbf{J}_f$	Jacobian matrix
$\mathbf{Q} \succ 0$	Covariance matrix	$\mathbf{R} \succ 0$	Measurement noise
$\mathbf{G}_t$	Gain matrix	$\hat{x}_{est,t}^-, \hat{x}_{est,t}^+$	Prior / posterior state estimations
$\mathbf{W}_{Q_i}, \mathbf{W}_{F_i}, \mathbf{W}_{G_i}, \mathbf{W}_{R_i}$	Weight matrix	$\ \Xi\ _{\mathbf{W}} = \Xi^T \mathbf{W} \Xi$	Algebraic operations
$\mathbf{y}_{i,k t}^p$	Predictive output	$k t$	k -th time step
$\mathbf{y}_{i,k t}^{des}$	Desired output	$\mathbf{W} \succeq 0$	Semi-positive definite
$\tilde{d}_{i,j}$	Desired interval distance	$ q_i $	Cardinality of q_i
$\mathfrak{A}$	Adjacency matrix	$\mathfrak{D}$	In-degree matrix
$\mathfrak{P}$	Pinning matrix	J_i^*	Cost function
T_i^t	Driving/braking torque	$f_{veh,i}$	Vehicle resistance
$C_{aero,i}$	Air drag coefficient	g	Gravity coefficient
$f_{roll,i}$	Rolling resistance coefficient	$\otimes$	Kronecker product
$\mathbf{P}_t^-$	Priori covariance matrix	z_t	Leader's planing status

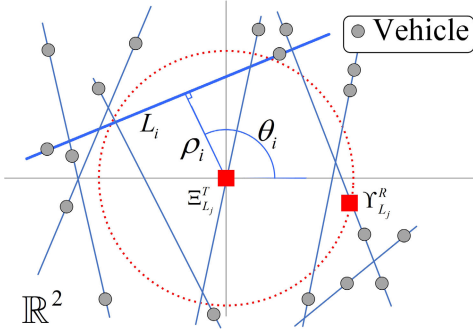


Fig. 2. V2V vehicular network model.

Definition 3 ([30]): (Motion invariant). A line process is said to be motion invariant if its distribution remains invariant during translation and rotation.

Consider a line process ψ_l characterized by a motion invariant property and a line density of μ_l . In C , the two-dimensional (2D) PPP is described by a line with a density λ_l , represented as ψ_C [31]. The relationship between λ_l and μ_l can be expressed as $\lambda_l = \mu_l/\pi$.

A. Wireless Node Model

The network [32] in a vehicle platoon consists of vehicle on-board unit (OBU) nodes for V2V communication. In this research, we focus only on V2V communication, excluding roadside units (RSUs) and macro base station nodes (MBSs), as shown in Fig. 1. Since the vehicles are positioned on a road, the irregular road space is modeled using a PLP

with a line density of μ_l [31]. The transmitting and receiving node positions are represented by one-dimensional (1D) PPP [31], denoted as $\Xi_{L_j}^T$ and $\Upsilon_{L_j}^R$, with densities λ_t and λ_r , respectively. Thus, the positions of vehicle transmitting nodes and vehicle receiving nodes form Cox processes [32] $\Phi_t = \Xi_{L_j}^T$ and $\Phi_r = \Upsilon_{L_j}^R$, respectively.

The wireless node model analyzes the probability of V2X signal coverage based on the signal interference ratio (SIR) at a receiver node, randomly selected from the vehicle node PPP. To simplify, we shift the receiver's position to the origin $o := (0, 0)$, and assume it lies on the road L_0 . Thus, the line process becomes $\Phi_{l_0} := \Phi_l \cup L_0$. Similarly, the PPP for the receiver node is $\Phi_{r_0} := \Phi_r \cup \Upsilon_{L_0}^R \cup \{o\}$, representing a collection of 1D PPPs with density λ_r along L_0 . The transmitting node's 1D PPP is $\Phi_{t_0} := \Phi_t \cup \Upsilon_{L_0}^V$, representing a set of 1D PPPs with line density λ_t on L_0 .

The transmitting power of the vehicle node is denoted as P_t . Since communication occurs between vehicle nodes and receiver nodes along a road, the vehicle transmitter enhances the signal at the receiver through beamforming. We assume V2V communication uses a sector antenna, with main lobe and side lobe gains represented as G_t and g_t , respectively. The vehicle node positions form a Cox process Φ_v , realized by a line Φ_{l_0} , according to the Palm probability of the receiving point process. The parameters $\{P_t, G_t, g_t\}$ represent the transmit power, main lobe gain, and side lobe gain, respectively.

The channel fading characteristics can greatly affect the communication quality and the success rate of information transmission. Therefore, the model of Nakagami-m is chosen to simulate the channel fading. Recognizing the significant variation in fading parameters and channel gains across different roads in the same line-of-sight (LoS) environment, we define the channel fading parameters on and off the L_0 as Pa_{l_0} and $Pa_{\tilde{l}_0}$. Similarly, the fading gains on and off the L_0 are denoted as Ga_{l_0} and $Ga_{\tilde{l}_0}$. The fading gain Ga_i , where $i \in \{l_0, \tilde{l}_0\}$ follows a Gamma distribution with probability density function (PDF) [31]:

$$f_{Ga_i}(\hat{g}) = \frac{(Pa_i)^{(Pa_i)} g^{(Pa_i)-1}}{\Gamma(Pa_i)} e^{-(Pa_i)\hat{g}}, \quad Pa_i \in \{Pa_{l_0}, Pa_{\tilde{l}_0}\} \quad (1)$$

The path loss of V2V communication link is assumed to be follow an exponential $\alpha > 2$ standard power-law model. We denote the shadowing effects for the links from the typical receiving node to the transmitting node on the typical line as χ_{l_0} , as well as on the atypical line as $\chi_{\tilde{l}_0}$. Let χ_i , where $i \in \{l_0, \tilde{l}_0\}$ follows a log-normal distribution as $10 \log_{10} \chi_i \sim N(w_i, \sigma_i^2)$, where w_i is the mean and σ_i is the standard deviation in dB. Therefore, the received power by a typical receiving node from a transmitting node at location x is denoted as [31] and [32]:

$$P_r(x) = \begin{cases} P_t G_t Ga_{l_0} \chi_{l_0} \|x\|^{-\alpha}, & x \in \Xi_{L_0} \\ P_t g_t Ga_{\tilde{l}_0} \chi_{\tilde{l}_0} \|x\|^{-\alpha}, & x \in \Phi_t \setminus \Xi_{L_0} \end{cases} \quad (2)$$

To assess the success rate of information acquisition, we assume a typical receiving node gets valid data from

the strongest transmitting node. However, in a platoon, multiple nodes may transmit simultaneously. To address this, we establish the relationship between the information acquisition success rate and communication distance, allowing calculation of the success rate for various transmitting nodes based on distance.

The interference at a typical receiving node comprises contributions from the typical line node (I_{l0}) and other line nodes ($I_{\tilde{l}0}$). Consequently, the SIR is expressed as [32]:

$$\text{SIR} = \frac{P_r(x)}{I_{l0} + I_{\tilde{l}0}} \quad (3)$$

$$I_{l0} = \sum_{\substack{x \in \Xi_{L0} \\ P_t G_t G_{a_{l0}} \chi_{l0} \|x\|^{-\alpha} < P_r(x^*)}} P_t G_t G_{a_{l0}} \chi_{l0} \|x\|^{-\alpha} \quad (4)$$

$$I_{\tilde{l}0} = \sum_{L_j \in \{\Phi_{l0} \setminus L_0\}} \sum_{x \in \Xi_{L_j}} P_t g_t G_{a_{\tilde{l}0}} \chi_{\tilde{l}0} \|x\|^{-\alpha} \quad (5)$$

where, $x^* = \arg\max_{x \in \Xi_{L0}} P_t G_t G_{a_{l0}} \chi_{l0} \|x\|^{-\alpha}$.

Assuming that V2V is the only communication method available and no other means of information exchange exist within the platoon, let ε denote this communication event. The probability of ε occurring is given by $\mathbb{P}(\varepsilon) = 1$. The PDF and CDF(cumulative distribution function) for the distance between a typical receiving node and the nearest communication node are given by [32]:

$$\text{CDF} : F_{dis}(r) = 1 - \exp(-\lambda_t 2r) \quad (6)$$

$$\text{PDF} : f_{dis}(r) = 2\lambda_t \exp(-\lambda_t 2r) \quad (7)$$

Lemma1. The Laplace transform of an interfering signal at a typical receiving node, subject to event ε occurrence and communication service distance R as:

$$\begin{aligned} \mathcal{L}_{I|R,\varepsilon}(s) &= \exp(2\lambda_t \int_r^\infty (1 - (1 + \frac{P_t G_t \|r\|^{-\alpha}}{G_{a_{l0}}})^{-\alpha}) dr \\ &\quad - 2\pi \lambda_t \int_0^\infty (1 - (1 + \frac{P_t G_t \|r\|^{-\alpha}}{G_{a_{\tilde{l}0}}})^{-\alpha}) r dr) \end{aligned} \quad (8)$$

Proof: In a stochastic process, when the individual processes are independent, the Laplace transform of their sum equal the product of the Laplace transforms of each process. In this paper, since the interfering signals I_{l0} and $I_{\tilde{l}0}$ are independent of each other, the Laplace transform of their sum of signal interference can be simplified as follows [31], [32]:

$$\mathcal{L}_{I|R,\varepsilon}(s) = \mathcal{L}_{I_{l0}|R,\varepsilon}(s) \mathcal{L}_{I_{\tilde{l}0}|R,\varepsilon}(s) \quad (9)$$

For $G_{a_{l0}} \sim \text{Gamma}(k, \theta)$, the eigenfunctions are:

$$\mathbb{E}(e^{-t G_{a_{l0}}}) = (1 + \zeta t)^{-k} \quad (10)$$

where k denotes the shape parameter and ζ is the scale parameter. The Laplace transform of the PPP is expressed as:

$$\mathbb{E}[\prod_{x \in \Phi_i} f(x)] = \exp(-\lambda \int_{R/R^2} [1 - f(x)] dx) \quad (11)$$

The received interference at typical receiver from nodes on the L_0 is modeled as a linear 1D PPP. Therefore, its Laplace transform is derived as follows:

$$\begin{aligned} \mathcal{L}_{I_{l0}|R,\varepsilon} &= \mathbb{E}[\exp(-s I_{l0})] \\ &= \mathbb{E}[\exp(-s \sum_{x \in \Phi_{l0}} P_t G_t G_{a_{l0}} \|x\|^{-\alpha})] \\ &\stackrel{a}{=} \mathbb{E}_{x \in \Phi_{l0}} \left[\prod_{x \in \Phi_{l0}} \mathbb{E}_{G_{a_{l0}}} [\exp(-s P_t G_t G_{a_{l0}} \|x\|^{-\alpha})] \right] \\ &\stackrel{b}{=} \mathbb{E}_{x \in \Phi_{l0}} \left[\prod_{x \in \Phi_{l0}} (1 + \frac{s P_t G_t \|x\|^{-\alpha}}{P a_{l0}})^{-P a_{l0}} \right] \\ &\stackrel{c}{=} \exp \left[-\lambda_t \int_0^\infty (1 - (1 + \frac{s P_t G_t \|x\|^{-\alpha}}{P a_{l0}})^{-P a_{l0}}) dx \right] \end{aligned} \quad (12)$$

where, step *a* is a process where the interference from each other are independent, step *b* involves a process that should follow a Gamma distribution. Finally, step *c* deals with the Laplace transform of a 1D PPP. Considering only nodes excluding the L_0 line, the interference can be modeled as a 2D PPP. Consequently, its Laplace transform can be derived as follows:

$$\begin{aligned} \mathcal{L}_{I_{\tilde{l}0}|R,\varepsilon} &= \mathbb{E}[\exp(-s I_{\tilde{l}0})] \\ &= \mathbb{E}[\exp(-s \sum_{x \in \Phi_{\tilde{l}0}} P_t G_t G_{a_{\tilde{l}0}} \|x\|^{-\alpha})] \\ &\stackrel{a}{=} \mathbb{E}_{x \in \Phi_{\tilde{l}0}} \left[\prod_{x \in \Phi_{\tilde{l}0}} \mathbb{E}_{G_{a_{\tilde{l}0}}} [\exp(-s P_t G_t G_{a_{\tilde{l}0}} \|x\|^{-\alpha})] \right] \\ &\stackrel{b}{=} \mathbb{E}_{x \in \Phi_{\tilde{l}0}} \left[\prod_{x \in \Phi_{\tilde{l}0}} (1 + \frac{s P_t G_t \|x\|^{-\alpha}}{P a_{\tilde{l}0}})^{-P a_{\tilde{l}0}} \right] \\ &\stackrel{c}{=} \exp[-\lambda_t \int_0^\infty \int_0^{2\pi} (1 - (1 + \frac{s P_t G_t \|r\|^{-\alpha}}{P a_{\tilde{l}0}})^{-P a_{\tilde{l}0}}) r dr d\theta] \\ &\stackrel{d}{=} \exp[-2\pi \lambda_t \int_0^\infty (1 - (1 + \frac{s P_t G_t \|r\|^{-\alpha}}{P a_{\tilde{l}0}})^{-P a_{\tilde{l}0}}) r dr] \end{aligned} \quad (13)$$

where independent interference between nodes leads to step *a*, while step *b* follows a Gamma distribution. Step *c* involves the Laplace transform of 2D PPP, converting the integral process into polar coordinates. The final equation is obtained in step *d* through integration θ . (8) is derived by substituting (12) and (13) into (9).

B. Successful Probability of Message Transmission

The success probability of message transmission is defined as the probability that the SIR at a receiving node exceeds a threshold β . Using the Laplace transform of conditional distributions, we derive this probability for V2V communication, considering service distance and interference power. This probability assesses platoon network connectivity and guides corrective measures for transmitted

data. The success probability at a typical receiving node is given by:

$$\mathbb{P}_{succ} = \sum_{k=0}^{Pa-1} \int_0^\infty \left[\frac{\left(\frac{Pa\beta r^\alpha I}{P_t G_t}\right)^k}{k!} \left[\frac{\partial^k}{\partial s^k} \mathcal{L}_{I|R,\varepsilon}(s) \right]_{s=\frac{Pa\beta r^\alpha I}{P_t G_t}} \right] f_{dis}(r) dr \quad (14)$$

Proof: Following the methodology of literature [32], [33], we derive conditional probabilities for the success rate of data acquisition at typical receiver nodes as:

$$\begin{aligned} \mathbb{P}_{succ} &= E_I[\mathbb{P}(\text{SIR} > \beta | R, x)] \\ &\stackrel{a}{=} E_I[\mathbb{P}(Ga > \frac{\beta I}{P_t G_t \chi r^{-\alpha}} | R, x)] \\ &\stackrel{b}{=} E_I\left[\frac{\Gamma(Pa, \frac{Pa\beta r^\alpha I}{P_t G_t})}{\Gamma(Pa)} | R, x\right] \\ &\stackrel{c}{=} E_I\left[\sum_{k=0}^{Pa-1} \frac{\left(\frac{Pa\beta r^\alpha I}{P_t G_t}\right)^k}{k!} \exp\left(-\frac{Pa\beta r^\alpha I}{P_t G_t}\right) | R, x\right] \\ &\stackrel{d}{=} \sum_{k=0}^{Pa-1} \int_0^\infty \left[\frac{\left(\frac{Pa\beta r^\alpha I}{P_t G_t}\right)^k}{k!} \exp\left(-\frac{Pa\beta r^\alpha I}{P_t G_t}\right) | R, x \right] f_{dis}(r) dr \\ &\stackrel{e}{=} \sum_{k=0}^{Pa-1} \int_0^\infty \left[\frac{\left(\frac{Pa\beta r^\alpha I}{P_t G_t}\right)^k}{k!} \left[\frac{\partial^k}{\partial s^k} \mathcal{L}_{I|R,\varepsilon}(s) \right]_{s=\frac{Pa\beta r^\alpha I}{P_t G_t}} \right] f_{dis}(r) dr \end{aligned} \quad (15)$$

Equation (15) derives the success probability of data transmission. The calculation integrates SIR using process a , applies the Gamma CCDF in process b , and incorporates the incomplete Gamma function in process c . Process d uses the expectation definition, while process e leverages the relationship between exponential expressions and Laplace transforms.

C. Network Connectivity

The decision to use real-time data from neighboring vehicles or estimated data relies on the network status indicator. Network connectivity serves as the key criterion. During interruptions, the controller uses estimated data as input. For a communication network G , $V(G) = \{v_1, v_2, \dots\}$ and $E(G) = \{e_1, e_2, \dots\}$ denote the set of all nodes and edges in G , respectively. Then the network G can be expressed as $G = (V, E)$. If we denote by $n = |V(G)|$ the number of nodes in $V(G)$ and by $m = |E(G)|$ the number of edges in $E(G)$, then the network $G = (n, m)$. Form the n -order neighbor matrix $A_{nei}(G)$ of the network G with the element $a_{ij}(i, j = 1, 2, \dots)$, where $a_{ij} = 1$ indicates that the node v_i is adjacent to the node v_j , and $a_{ij} = 0$ indicates that the node v_i is not adjacent to the node v_j .

In a vehicle platoon with a directed communication network, connectivity between nodes can be maintained via directed paths, even without bidirectional links. The deterministic

method first converts the directed graph into an undirected one, then applies connectivity assessment for undirected networks.

Definition 4: Given an undirected network G of order $n(n > 2)$, and its set of neighbors $A_i(i = 1, 2, \dots)$, the algorithm for set of neighbors is as follows:

$$A_i \cap A_j = \{z_k = x_k \cap y_k | x_k \in A_i, y_k \in A_j, k = 1, 2, \dots, n\} \quad (16)$$

$$A_i \cup A_j = \{z_k = x_k \cup y_k | x_k \in A_i, y_k \in A_j, k = 1, 2, \dots, n\} \quad (17)$$

The determination of the connectivity of an n -order undirected network G is as follows:

- i) Generate the neighbor matrix $A_{nei}(a_{ij}) \in \mathbb{R}^{n \times n}$ and obtain the set of neighbors A_i .
- ii) Find $A_* \cap A_j$ to determine whether there is a common neighbor between the two neighbor sets, where $A_* = A_1$. If not, then increment i by 1 and continue to find the intersection. If there is a common neighbor, find $A_* = A_* \cup A_j$ and then increment i by 1.
- iii) At the end of the above iteration, if $|A_*| < n$, then the network is not connected. Otherwise, it can be determined that the network is connected.

To demonstrate the effectiveness of our proposed method, we compared it with [34]. We applied a connectivity determination approach and analyzed the time complexity upper bounds. Our method has a time complexity of $O(3n^2)$, while the method in [34] has $O(2n^3)$. This comparison emphasizes the computational efficiency of our approach.

III. PLATOON MODELS AND VEHICLE STATE ESTIMATION

A. Non-Linear Platoon Modeling

In this study, only the longitudinal dynamics model of a vehicle is considered, which includes the engine, driveline system, braking system, air resistance, and rolling resistance. To balance model accuracy and brevity, we make the following assumptions: i) The vehicles platoon traveled on a straight roadway. ii) The vehicle's driving and braking torque are controlled by a single uniform control variable. Thus, vehicle i is modeled as [35] and [36]:

$$\begin{cases} p_i^{t+1} = p_i^t + v_i^t \Delta t \\ v_i^{t+1} = v_i^t + \frac{\Delta t}{m_{veh,i}} \left(\frac{\eta_{veh,i}}{R_{rad,i}} T_i^t - f_{veh,i}(v_i^t) \right) \\ T_i^{t+1} = T_i^t - \frac{1}{\tau_i} T_i^t \Delta t + \frac{1}{\tau_i} u_i(t) \Delta t \\ f_{veh,i}(v_i^t) = C_{aero,i}(v_i^t)^2 + m_{veh,i} g f_{roll,i} \end{cases} \quad (18)$$

The nonlinear dynamics model of the vehicle can be simplified into the following state-space equations:

$$\begin{aligned} \mathbf{x}_i(t+1) &= \mathbf{A}_i \mathbf{x}_i(t) + \mathbf{B}_i \mathbf{u}_i(t) + \mathbf{w}_i(t) \\ \mathbf{y}_i(t) &= \mathbf{C} \mathbf{x}_i(t) \end{aligned} \quad (19)$$

$$\mathbf{x}_i(t) = \begin{bmatrix} p_i^t \\ v_i^t \\ T_i^t \end{bmatrix}, \mathbf{u}_i(t) = u_i(t), \mathbf{C} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad (20)$$

$$\mathbf{A}_i = \begin{bmatrix} 1 & \Delta t & 0 \\ 0 & 1 & \frac{\Delta t \eta_{\text{veh},i}}{m_{\text{veh},i} R_{\text{rad},i}} \\ 0 & 0 & 1 - \frac{\Delta t}{\tau_i} \end{bmatrix}, \mathbf{B}_i = \begin{bmatrix} 0 \\ 0 \\ \frac{\Delta t}{\tau_i} \end{bmatrix} \quad (21)$$

$$\mathbf{w}_i(t) = \begin{bmatrix} 0 \\ -\frac{\Delta t}{m_{\text{veh},i}} (C_{\text{aero},i} (v_i^t)^2 + m_{\text{veh},i} g f_{\text{roll},i}) \\ 0 \end{bmatrix} \quad (22)$$

B. Estimation of V2V Message Under Dynamic Networks

Maintaining platoon stability in dynamic networks requires estimating the state of the leader or preceding vehicles. This estimation tracks trajectories for vehicles missing V2V messages. As leader vehicle information critically affects platoon stability, this paper focuses on estimating and correcting its status information to ensure platoon stability [35], as demonstrated in the simulations. An extended Kalman filter (EKF) is employed to ensure precise state estimation [37]. To improve the stability of the estimation method, the communication success rate is leveraged to balance the weight between prior predictions and observations. The estimated state of the leader vehicle, $\mathbf{x}_{\text{est}} = [p, v]^\top$, is determined as follows:

$$\widehat{\mathbf{x}}_{\text{est},t} = f(\widehat{\mathbf{x}}_{\text{est},t-1}, u_t) = \mathbf{x}_{[1:2,1:2]} \quad (23)$$

$$\mathbf{P}_t^- = \mathbf{J}_f(\widehat{\mathbf{x}}_{t-1}) \mathbf{P}_{t-1} \mathbf{J}_f(\widehat{\mathbf{x}}_{t-1})^\top + \mathbf{Q}_{t-1} \quad (24)$$

$$\mathbf{J}_f = \begin{bmatrix} \frac{\partial f_1}{\partial p} & \frac{\partial f_1}{\partial v} \\ \frac{\partial f_2}{\partial p} & \frac{\partial f_2}{\partial v} \end{bmatrix} = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 - \frac{2C_{\text{aero}} v_t \Delta t}{m_{\text{veh}}} \end{bmatrix} [\widehat{\mathbf{x}}_t]_{t-1} \quad (25)$$

$$\mathbf{G}_t = \mathbf{P}_t^- \mathbf{H}^\top (\mathbf{H} \mathbf{P}_t^- \mathbf{H}^\top + \mathbf{R}_k)^{-1} \quad (26)$$

$$\widehat{\mathbf{x}}_{\text{est},t} = \widehat{\mathbf{x}}_{\text{est},t}^- + \mathbb{P}_{\text{succ}} \mathbf{G}_t (z_t - \mathbf{H} \widehat{\mathbf{x}}_{\text{est},t}^-) \quad (27)$$

$$\mathbf{P}_t = \mathbb{P}_{\text{succ}} (\mathbf{I} - \mathbf{G}_t \mathbf{H}) \mathbf{P}_t^- + (1 - \mathbb{P}_{\text{succ}}) \mathbf{P}_t^- \quad (28)$$

$$\mathbf{H} = [1, 0; 0, 1] \quad (29)$$

where, f corresponds to the first two terms of (18), and $\mathbf{P}_t^-$ and $\mathbf{P}_t$ are the priori and system error covariance matrices. $\mathbf{Q}$ and $\mathbf{R}$ represent the covariance matrices for the system process noise and measurement noise, respectively. $\mathbf{G}$ is the gain matrix, and $\widehat{\mathbf{x}}_{\text{est},t}^-$ and $\widehat{\mathbf{x}}_{\text{est},t}$ denote the prior and posterior estimations of the state information. z_t is the observation state of the leader vehicle, specifically its position and speed. When the network is connected, the observation state is transmitted to the follower vehicles. In the case of network interruptions, z_t can be estimated using the leader vehicle's dynamic model $f(\widehat{\mathbf{x}}_{\text{est},t-1}, u_{t-1})$. $\mathbb{P}_{\text{succ}}$ acts as a weight between the estimated and observed states. When the network is reliable, the estimator relies more on the new measurements. Conversely, when $\mathbb{P}_{\text{succ}}$ is lower, the estimator places greater weight on the predicted state. This weight is dynamically adjusted based on network performance, enhancing the sensitivity of the estimation model.

The state of the leader vehicle is estimated when a vehicle lost V2V communication. The following vehicles that have lost V2V communication are controlled within the platoon based on the estimated state. Once the communication is restored, the system error covariance matrix and gain matrix are updated using the state information obtained from the V2V message.

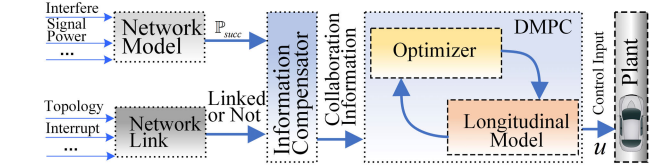


Fig. 3. The structure of the control system.

IV. PLATOON CONTROL SYSTEMS UNDER DYNAMIC NETWORKS

Fig. 3 illustrates the control process for vehicle platooning in dynamic networks. Network performance and connectivity results are fed into the information compensator. The compensator assesses network conditions to determine whether to apply data compensation or update estimation model parameters. Collaborative information from the compensator is then used by the distributed MPC to compute optimal control inputs for the platoon.

A. Distributed Model Predictive Control of Platoon

Vehicle platooning, as a typical multi-agent system, benefits from DMPC, which can address multi-objective and multi-constraint problems. DMPC is capable of solving multiple optimization problems at each sampling time, making it well-suited for scenarios involving interactions among multiple vehicles. In this approach, each local controller makes decisions based on its own model, yet they coordinate to optimize the system's overall performance. The primary control goal is to ensure the follower vehicle matches the speed of the lead vehicle while maintaining a small desired gap. This goal informs the design of the optimization controller [27].

$$\lim_{t \rightarrow \infty} |v_i^t - v_0^t| \rightarrow 0, \lim_{t \rightarrow \infty} |p_i^t - p_j^t - \tilde{d}_{i,j}| \rightarrow 0 \quad (30)$$

The prediction horizon is set to $H^p = 20$, and the optimized control sequence for MPC is defined as

$$\mathcal{U}_{\text{opt},i}^k = (u_{\text{opt},i}^0, u_{\text{opt},i}^1, \dots, u_{\text{opt},i}^{H^p-1}) \quad (31)$$

The optimization objective of the DMPC problem is to optimize the total cost of each vehicle node over the defined time horizon.

$$\begin{aligned} J_i(e_{\text{des},i}, e_{u,i}, e_{\text{ter},i}, e_{i,j}) \\ = \sum_{k=0}^{H^p-1} \left\{ \|e_{\text{des},i}\|_{\mathbf{w}_{Q,i}} + \|e_{u,i}\|_{\mathbf{w}_{R,i}} \right. \\ \left. + \|e_{\text{ter},i}\|_{\mathbf{w}_{F,i}} + \|e_{i,j}\|_{\mathbf{w}_{G,i}} \right\} \end{aligned} \quad (32)$$

Subject to:

$$\mathbf{y}_i^p(k|t) = \mathbf{C} \mathbf{x}_i^p(k|t) \quad (33)$$

$$k = 0, 1, \dots, H^p - 1 \quad (34)$$

$$\mathbf{x}_i^p(0|t) = \mathbf{x}_i^p(t) \quad (35)$$

$$u_{i,t}^k \in [u_{\min,i}, u_{\max,i}] \quad (36)$$

$$\mathbf{y}_{i,H^p|t}^p = 1/|Q_i| \times \sum_{j \in Q} (\mathbf{y}_{j,H^p|t}^a + \tilde{d}_{i,j}) \quad (37)$$

$$\mathbf{y}_{i,H^p|t+1}^a = (\mathbf{I} + \mathcal{D} \cdot \Delta t) \mathbf{y}_{i,H^p|t}^p, \quad \mathcal{D} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad (38)$$

The first term in (32), $e_{des,i}$, captures the deviation between the predicted output sequence (vehicle speed and position) $\mathbf{y}_{i,k|t}^p$ and the desired output state $\mathbf{y}_{i,k|t}^{des}$. The second term, $e_{u,i}$, represents the discrepancy between the control input $u_{i,k|t}$ and the vehicle's resistance torque $f_{veh,i}$, aiming to limit excessive acceleration and deceleration:

$$e_{des,i} = \mathbf{y}_{i,k|t}^p - \mathbf{y}_{i,k|t}^{des} \quad (39)$$

$$\mathbf{y}_{i,k|t}^{des} = [p_{0,(H^p|t)} - i \cdot d, v_{0,t}]^\top \quad (40)$$

$$e_{u,i} = u_{i,k|t} - f_{veh,i}(v_{i,k|t}^p) \quad (41)$$

In (32), the term $e_{ter,i}$ constrains vehicle i to follow the assumed trajectory $\mathbf{y}_{i,k|t}^a$ of other nodes. Meanwhile, $e_{i,j}$ ensures that vehicle i follows the assumed trajectory $\mathbf{y}_{j,k|t}^a$ of its neighboring vehicle j , where $\tilde{d}_{i,j} = d \times |j - i|$:

$$e_{ter,i} = \mathbf{y}_{i,k|t}^p - \mathbf{y}_{i,k|t}^a \quad (42)$$

$$e_{i,j} = \mathbf{y}_{i,k|t}^p - \mathbf{y}_{j,k|t}^a + \tilde{d}_{i,j} \quad (43)$$

where $\mathbf{y}_{i,k|t}^a$ represents the assumed output, iteratively calculated based on the assumed control input using equations (18) and (19). The weight matrix $\mathbf{W}_{Q,i}$ constrains the desired output, while $\mathbf{W}_{R,i}$ controls the deviation of the control variables. $\mathbf{W}_{F_i}$ is the weight matrix that constrains the error between the desired and assumed outputs, and $\mathbf{W}_{G_i}$ limits the deviation between neighboring vehicles. Equation (33) computes the predicted output using (18). The terminal prediction is determined by (37), and the control variable constraints are defined by (36). To ensure the feasibility and stability of the control within the objective function, we introduce the terminal constraints along with the assumption error. Specifically, $\mathbf{x}_{i,t+1}^0 = \mathbf{x}_{i,t}^{H^p}$ represents the terminal constraints of the state variable, indicating that the terminal state value $\mathbf{x}_{i,t}^{H^p}$ at time t serves as the initial value $\mathbf{x}_{i,t+1}^0$ at step 0 for time $t + 1$. The assumption error is from (42) based on assumed control input, which is given by $u_{i,t+1}^{a,k} = u_{opt,i,t}^{k+1}$ for $k = 0, 1, \dots, H^p - 2$, and $u_{i,t+1}^{a,k} = f_{veh,i}(v_{opt,i,t}^{H^p})$ when $k = H^p - 1$ [35]. The constraint $u_{i,H^p-1|t+1}^a = f_{veh,i}(v_{opt,i,H^p|t})$ at $k = H^p - 1$ ensures that the assumed control variables align with the actual vehicle requirements, thereby maintaining platoon stability.

B. Stability Analysis of Platoon Control Under Dynamic Networks

The stability of distributed MPC algorithms is analyzed using a Lyapunov function based on the local cost function, showing its decreasing property. The analysis proceeds in two steps: first, demonstrating that predictive outputs converge to the desired outputs, and second, establishing the asymptotic decrease of the Lyapunov function. Step 1: If the network topology G is connected, the terminal output state within the predictive horizon will converge to the desired state. Thus, the error $e_{des,i}$ converges to zero.

$$\lim_{t \rightarrow \infty} |e_{des,i}| = 0 \quad (44)$$

Proof: For each vehicle node moving at a constant speed towards the time horizon's end, it follows from (33), (35), and (36) that:

$$e_{des,i,H^p|t+1} = \mathbf{y}_{i,H^p|t+1}^p - \mathbf{y}_{i,H^p|t+1}^{des}$$

$$\begin{aligned} &= \frac{(I + \mathcal{D} \cdot \Delta t)}{|Q_i|} \sum_{j \in Q_i} \mathbf{y}_{j,H^p|t}^p + \frac{1}{|Q_i|} \sum_{j \in Q_i} \tilde{d}_{i,j} - \mathbf{y}_{i,N^p|t+1}^{des} \\ &= \frac{(I + \mathcal{D} \cdot \Delta t)}{|Q_i|} \sum_{j \in Q_i} \mathbf{y}_{j,H^p|t}^p - \frac{(I + \mathcal{D} \cdot \Delta t)}{|Q_i|} \sum_{j \in Q_i} \mathbf{y}_{j,H^p|t}^{des} \\ &= \frac{(I + \mathcal{D} \cdot \Delta t)}{|Q_i|} \sum_{j \in Q_i} [\mathbf{y}_{j,H^p|t}^p - \mathbf{y}_{j,k|t}^{des}] \\ &= \frac{(I + \mathcal{D} \cdot \Delta t)}{|Q_i|} \sum_{j \in Q_i} e_{des,j,H^p|t} \end{aligned} \quad (45)$$

The error of various vehicles is presented as a collective set, given by $E_{des,H^p|t} = [e_{des,1,H^p|t}, \dots]^T \in \mathbb{R}^{2N \times 1}$. Therefore, (45) can be expressed as a collective set:

$$\begin{aligned} E_{des,H^p|t+1} &= [(\mathcal{D} + \mathfrak{P})^{-1} \cdot \mathfrak{A}] \\ &\quad \otimes (I + \mathcal{D} \cdot \Delta t) E_{des,H^p|t} \end{aligned} \quad (46)$$

Step 1: If the network is connected, the rank of G is n , meaning the topology matrix is nonsingular. From [35], it follows that the eigenvalues of $(\mathcal{D} + \mathfrak{P})^{-1} \cdot \mathfrak{A}$ lie within the unit circle. Since the eigenvalues of $(I + \mathcal{D} \cdot \Delta t)$ are all one, $E_{des,H^p|t}$ asymptotically converges to zero, proving step 1.

Step 2: We prove the stability of the system under network interruptions, assuming that the estimation error remains bounded. Let $V(t) = J_i^*(t)$ be the Lyapunov function candidate and $E(t)$ be the platoon error vector defined relative to the true leader state $x_0(t)$.

Assumption 1: The EKF estimation error $\epsilon(t)$ and the process noise $w(t)$ are uniformly bounded. There exist constants $B_\epsilon > 0$ and $B_W > 0$ such that $\|\epsilon(t)\| \leq B_\epsilon$ and $\|w(t)\| \leq B_W$ for all $t \geq 0$.

Assumption 2 ([38]): Nominal Decrease: If $\epsilon(t) = 0$ and $w(t) = 0$, there exists a class- $\mathcal{K}$ function $\alpha(\|E\|)$ ($\alpha(s) = \alpha_0 \|s\|^2$) such that for the ideal system trajectory $V_{ideal}(t+1) - V(t) \leq -\alpha(\|E(t)\|)$. This decrease arises because the MPC optimization minimizes the cost over the prediction horizon. The optimal cost $V(x(t))$ is Lipschitz continuous with respect to the state $x(t)$. There exists Lipschitz constant $L_V > 0$ such that $|V(x_a) - V(x_b)| \leq L_V \|x_a - x_b\|$. Lipschitz continuity of control law $u^*(t) = \mathcal{K}_{mpc}(x(t), x_{est,0}(t))$ is locally Lipschitz continuous with respect to the EKF estimation error $\epsilon(t)$. There exists Lipschitz constant $L_u > 0$ such that: $\|u^*(t) - u_{ideal}^*(t)\| \leq L_u \|\epsilon(t)\|$.

Actual state at $t + 1$: $x(t + 1) = f(x(t), u^*(t), w(t))$. Ideal state at $t + 1$: $x_{ideal}(t + 1) = f(x(t), u_{ideal}^*(t), 0)$. The deviation between the actual and ideal states is then given by:

$$\begin{aligned} &x(t + 1) - x_{ideal}(t + 1) \\ &= f(x(t), u^*(t), w(t)) - f(x(t), u_{ideal}^*(t), 0) \\ &= [Ax(t) + Bu^*(t) + w(t)] - [Ax(t) + Bu_{ideal}^*(t)] \\ &= B(u^*(t) - u_{ideal}^*(t)) + w(t) \\ &= B\Delta u(t) + w(t) \end{aligned} \quad (47)$$

The Lyapunov function difference is shown as

$$\begin{aligned} \Delta V(t) &= V(x(t + 1)) - V(x(t)) \\ &= V(x_{ideal}(t + 1)) - V(x(t)) \\ &\quad + V(x(t + 1)) - V(x_{ideal}(t + 1)) \end{aligned} \quad (48)$$

Using Lipschitz continuity (Assumptions 2): $|V(x(t+1)) - V(x_{\text{ideal}}(t+1))| \leq L_V(|B||L_u||\epsilon(t)| + |w(t)|)$. For the ideal case, nominal decrease ensures $V(x_{\text{ideal}}(t+1)) - V(x(t)) \leq -\alpha(|E(t)|)$. Combining the two terms

$$\Delta V(t) \leq -\alpha(|E(t)|) + L_V|B||L_u||\epsilon(t)| + L_V|w(t)| \quad (49)$$

There exist suitable class- $\mathcal{K}$ functions $\sigma_\epsilon(|\epsilon(t)|)$ and $\sigma_w(|w(t)|)$ such that

$$L_V|B||L_u||\epsilon(t)| + L_V|w(t)| \leq \sigma_\epsilon(|\epsilon(t)|) + \sigma_w(|w(t)|) \quad (50)$$

According to Assumption 1:

$$\Delta V(t) \leq -\alpha(|E(t)|) + \sigma_\epsilon(B_\epsilon) + \sigma_w(B_W) \leq -\alpha(|E(t)|) + B_{\text{total}}^2 \quad (51)$$

where $B_{\text{total}}^2 = \sigma_\epsilon(B_\epsilon) + \sigma_w(B_W)$. This implies that whenever $|E(t)|$ is large enough such that $\alpha(|E(t)|) > B_{\text{total}}^2$, then $\Delta V(t) < 0$, meaning the Lyapunov function decreases, thereby ensuring the stability of the platoon.

C. Platoon Performance Evaluation

This study evaluates the impact of communication on platoon performance using four metrics: The cumulative followability indicator (CFI) is defined as follows [27]:

$$\text{CFI} = \sum_{i=1}^N \frac{1}{T} \left[\sum_{t=0}^T (|W_v \times \Delta v_i^t| + |W_d \times \Delta d_i^t|) \right] \quad (52)$$

where T is the simulation length, Δv and Δd are the velocity and spacing errors, and $W_v = 20$, $W_d = 50$ denote the sensitivity to these errors.

The cumulative driving comfort (CDC) indicator, where a lower value indicates better comfort, is expressed as:

$$\text{CDC} = \sum_{i=1}^N \frac{1}{T} \left[\sum_{t=0}^T \left(\frac{d(a_i^t)}{dt} \right) \right] \quad (53)$$

The cumulative fuel consumption (CFC) metric evaluates the total energy consumed by all vehicles in the platoon. A lower CFC indicates better energy efficiency.

$$\text{CFC} = \sum_{i=1}^N \sum_{t=0}^T \begin{cases} \varepsilon_0 P_{\text{veh},i,t} + \varepsilon_1 P_{\text{veh},i,t}^2 + \varepsilon_2 P_{\text{veh},i,t}^3, & P_{\text{veh},i,t} \geq 0 \\ \varepsilon_0, & P_{\text{veh},i,t} < 0 \end{cases} \quad (54)$$

$$P_{\text{veh},i,t} = \left(\frac{f_{\text{veh},i}(v_i^t) + 1.04 m_{\text{veh},i} a_i^t}{3600 \eta_{\text{veh},i}} \right) v_i^t \quad (55)$$

where CFC is the cumulative fuel consumption, ε_i is the dimensionless coefficient, and $P_{\text{veh},i,t}$ is the instantaneous power of vehicle i at time t in kw.

The communication overhead metric measures the platoon's communication resource usage in a given topology. A higher value indicates greater resource demand.

$$\text{CO} = C_c \sum_{i=1}^N |\Pi_i| \quad (56)$$

where C_c represents the sensitivity of communication overhead, and $|\Pi_i|$ is the cardinality of set Π_i .

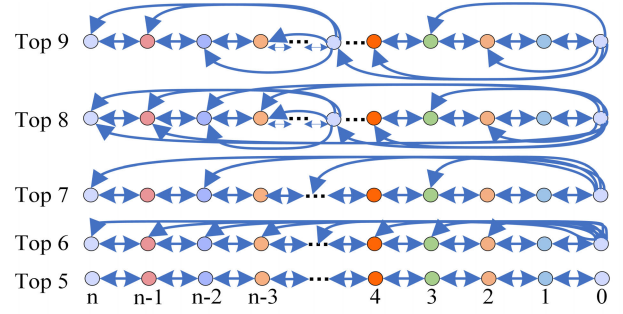


Fig. 4. Five newly added topologies.

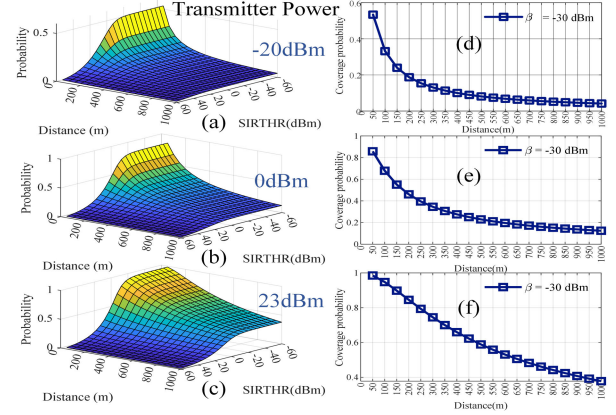


Fig. 5. Success probability of information transmission.

V. NUMERICAL SIMULATION AND ANALYSIS

This study conducts three case studies on a laptop with an Intel Core i9 processor and 16GB of RAM. Case 1 evaluates platoon performance under 9 topologies and 14 heterogeneous following vehicles. Case 2 examines performance with changes or loss in a vehicle's information acquisition path. Case 3 simulates stochastic intermittent network interruptions to assess the proposed controller's effectiveness.

A. Scenario Design

The leader vehicle starts at position $p_0^0 = 0$ m with velocity $v_0^0 = 20$ m/s. Its speeds are 20 m/s for $t \leq 1$ s, $20 + 2(t-1)$ m/s for $t \in (1, 2]$, and 22 m/s for $t > 2$ s. The sampling interval is $\Delta t = 0.1$ s, the prediction step is set to $H^P = 20$, and the desired spacing gap is 20 m. Network topologies include four traditional types (PF, PLF, TPF, TPLF) [35] and five new ones, as shown in Fig. 4. The vehicle system parameters used in the simulations are consistent with those in [35]. The parameters for state estimation are derived from equations (23) to (29). Optimization is performed using the *fmincon* toolbox, with a tolerance of 1×10^{-5} and a maximum of 2000 iterations.

B. Network Performance Simulation and Analysis

In the V2V network simulation, line density is $\mu_l = 35$ [1/km²], node density is $\lambda_v = 15$ [nodes/km], fading parameters are $P_{a10} = \tilde{P}_{a10} = 1$, and the path loss index is $\alpha = 4$. Fig. 5 presents the results, highlighting that signal transmit power, communication distance, and received power threshold significantly influence communication success probability.

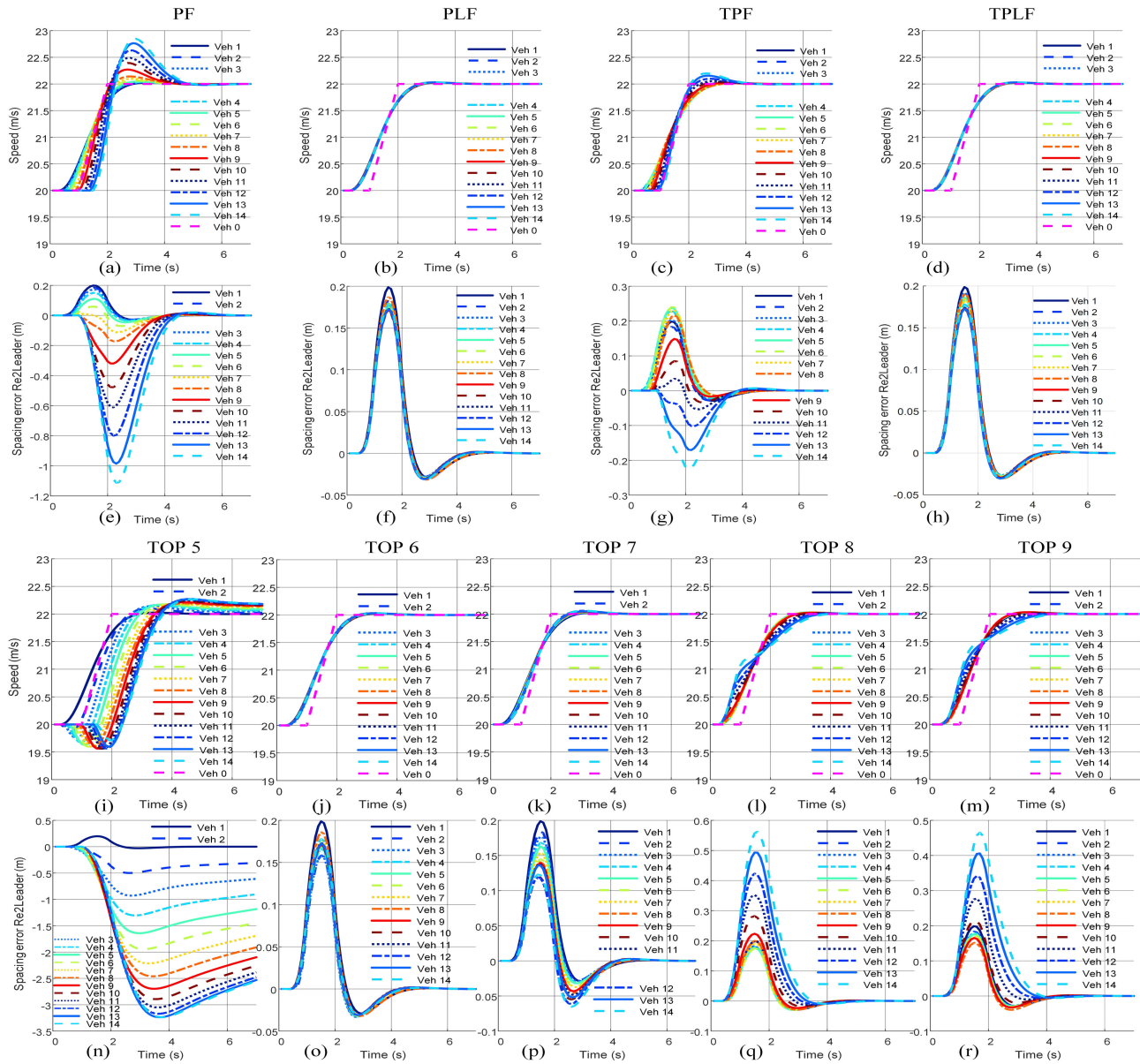


Fig. 6. Tracking performance of following vehicles.

Fig. 5 shows that transmission success probability decreases with longer communication distances but increases with lower received power thresholds. Results in Fig. 5(d)-(f) demonstrate that higher V2V transmit power improves reception success when the power threshold is $\beta = -30$ dBm. Based on 3GPP standards, this study sets transmit power at 23 dBm and the reception threshold at -30 dBm.

C. Analysis of Platoon Under 9 Network Topologies

This subsection evaluates a 15-vehicle platoon under nine network topologies, with results in Table II. Fig. 6 illustrates speed tracking (top) and spacing errors (bottom) relative to the leader (Veh 0). The CFI metric, based on spacing and speed errors, indicates better tracking with leader information (PLF, TPLF, TOP6). Tracking worsens with additional front and rear data, as in TOP5 (CFI 109.11). TOP7 achieves the

TABLE II
PLATOON PERFORMANCE UNDER MULTIPLE TOPOLOGIES

Network	CFI	CDC	CFC	CO
PF	9.08	0.84	14.83	33.60
PLF	2.67	0.50	12.52	64.80
TPF	2.86	0.65	13.15	64.80
TPLF	2.72	0.50	12.50	93.60
TOP-5	109.11	0.98	14.50	64.80
TOP-6	2.62	0.51	12.56	96.00
TOP-7	2.34	0.52	12.64	48.00
TOP-8	5.03	0.54	12.37	60.00
TOP-9	4.22	0.55	12.53	60.00

lowest CFI (2.34), balancing tracking accuracy with reduced communication overhead but slightly lower energy efficiency and comfort than PLF and TPLF.

Driving comfort, measured by CDC, depends on acceleration derivatives; lower values indicate better comfort. Table II

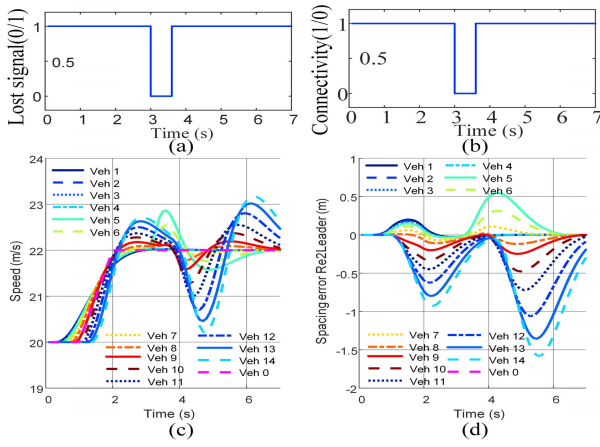


Fig. 7. Tracking performance of [35] under fixed-time outages.

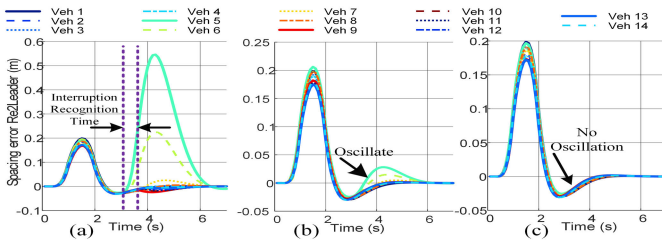


Fig. 8. Dynamic response under TPLF: (a) Control method in [17], (b) Proposed control method without consideration $\mathbb{P}_{succ}$, (c) Proposed control method with consideration $\mathbb{P}_{succ}$.

shows PLF and TPLF achieve the best comfort, with PLF improving tracking by 70.6% over PF. For energy consumption (CFC), TPLF and PLF perform best. Communication overhead is lowest in PF and highest in TOP6. This study prioritizes tracking performance, energy efficiency, and comfort, assigning less weight to communication overhead.

In summary: i) Topologies with leader vehicle information excel in tracking, comfort, and energy efficiency. ii) Bidirectional communication topologies perform worst in these metrics.

D. Platoon Performance Under Dynamic Network

This subsection simulates the proposed control method under network fluctuations, focusing on performance during outages. A case study examines Vehicle 5 under fixed-duration and random intermittent disconnections across various topologies. During a fixed communication outage (3-3.5 s), Vehicle 5 loses its V2V connections (Lost signal=0), disrupting platoon connectivity (Connectivity=0) (Fig. 7 a–b). First, the impact of message transmission success rate and network connectivity on platoon performance is analyzed using TPLF as an example. The timestamp-based network interruption detection method, which classifies a delay as an interruption when the waiting time T_r exceeds the threshold T_{thr} , introduces minor time lag, as shown in Fig.8 (a). This issue can be effectively mitigated by incorporating network connectivity. Additionally, as shown in Fig.8 (b), the control model without considering $\mathbb{P}_{succ}$ exhibits spacing error oscillations, leading to degraded platoon performance compared to the model that

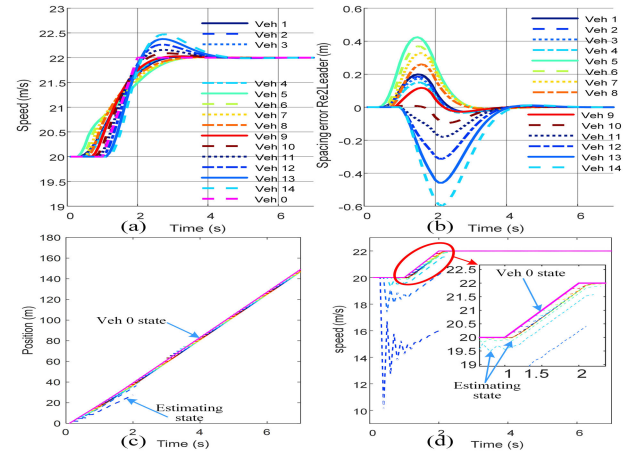


Fig. 9. Tracking performance of Prop under fixed-time outages.

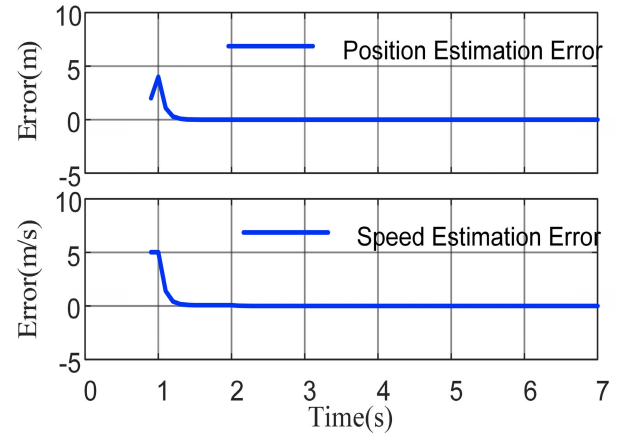


Fig. 10. Position and speed estimation errors.

accounts for $\mathbb{P}_{succ}$ (Fig.8 (c)). The traditional method described in [35] (Trad), under PF conditions (Fig. 7c–d), destabilizes the following vehicles. In contrast, the proposed method (Fig. 9) leverages leader state estimation, significantly reducing speed and spacing errors (Fig. 9a–b) and enhancing tracking performance. Fig. 9(c–d) presents the state estimation results for the future 2 seconds, showing the predicted position and velocity during the operation of the control system. Initially, accuracy is limited due to the lack of adaptive model parameter adjustments. However, as the simulation progresses, accuracy improves. Fig. 10 shows the position and velocity estimation errors at the 10th time step, with root mean square errors of 0.26 m and 0.65 m/s, respectively. The estimation errors remain near zero, thanks to the dynamic adjustment of model parameters. These small errors demonstrate the effectiveness of the estimation method.

As shown in Table III, the proposed method (Prop) outperforms both [17] and [35]. The table shows that the proposed method surpasses other topologies in tracking performance, driving comfort, and energy efficiency under PLF, TPLF, and TOP-6 communication topologies. The communication overhead is 62.22 for PLF, 88.62 for TPLF, and 93.42 for TOP-6. When multiple topologies are available, prioritizing PLF, followed by TPLF and TOP-6, is recommended to

TABLE III
PLATOON SIMULATION RESULTS DURING FIXED-TIME NETWORK
OUTAGE(↑ DENOTES PERFORMANCE IMPROVEMENT)

Method	Network	CFI	CDC	CFC	CO
[35]	PF	32.20	1.82	17.52	33.42
[17]		8.46	1.21	15.60	33.42
Prop		6.31↑	0.70↑	13.45↑	33.42
[35]	PLF	10.30	0.64	12.80	62.22
[17]		3.85	0.57	12.78	62.22
Prop		2.69↑	0.50↑	12.52↑	62.22
[35]	TPF	22.20	1.3	14.25	62.22
[17]		5.64	0.62	14.01	62.22
Prop		3.73↑	0.63↑	12.79↑	62.22
[35]	TPLF	10.18	0.71	12.92	88.62
[17]		4.68	0.69	12.55	88.62
Prop		2.87↑	0.50↑	12.48↑	88.62
[35]	TOP-5	365.51	3.04	24.94	64.62
[17]		236.21	2.12	19.47	64.62
Prop		109.08↑	0.955↑	14.61↑	64.62
[35]	TOP-6	13.03	0.77	13.00	93.42
[17]		3.98	0.64	12.86	93.42
Prop		2.67↑	0.50↑	12.54↑	93.42
[35]	TOP-7	9.76	0.81	13.08	45.42
[17]		3.72	0.59	12.77	45.42
Prop		3.58↑	0.51↑	12.51↑	45.42
[35]	TOP-8	12.70	0.66	12.65	57.42
[17]		6.32	0.58	12.45	57.42
Prop		5.13↑	0.54↑	12.35↑	57.42
[35]	TOP-9	13.22	1.14	13.58	57.42
[17]		5.36	0.78	12.87	57.42
Prop		4.17↑	0.55↑	12.56↑	57.42

optimize communication resource usage. This table also shows that fuel consumption is reduced by 23.2% with our proposed method compared to the baseline [35] and by 13.8% compared to the LSTM-based [17] approach under PF topology. Similar reductions in energy consumption are observed under other communication topologies. Thus, our proposed method demonstrates an advantage. Compared to [35], Prop improves tracking accuracy by 80%, driving comfort by 62%, and energy efficiency by 23%. These results highlight the proposed method's effectiveness in stabilizing the platoon during fixed-time communication outages.

A simulation of Vehicle 5 under random intermittent network disconnections, shown in Fig. 11, was conducted. Fig. 12 illustrates that the proposed method outperforms the control method described in [35] in terms of CFI, CDC, and CFC across 9 network types during random network disruptions in Vehicle 5. The results indicate that, in a PF network, the proposed method (Prop) improves tracking by 87.1%, driving comfort by 72.5%, and energy efficiency by 23.2%, compared to the method in [35]. Simulations with 50 vehicles under PLF show negligible spacing error (Fig.13), demonstrating scalability.

VI. EXPERIMENT

The experiments were conducted on a vehicle with a curb weight of 1725 kg, dimensions of 4820 × 1890 mm, a 2900 mm wheelbase, and a minimum turning radius of

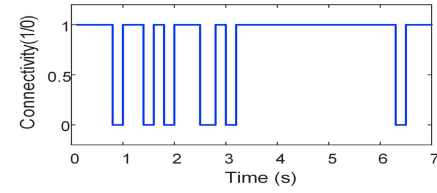


Fig. 11. Random Intermittent Network Disconnections.

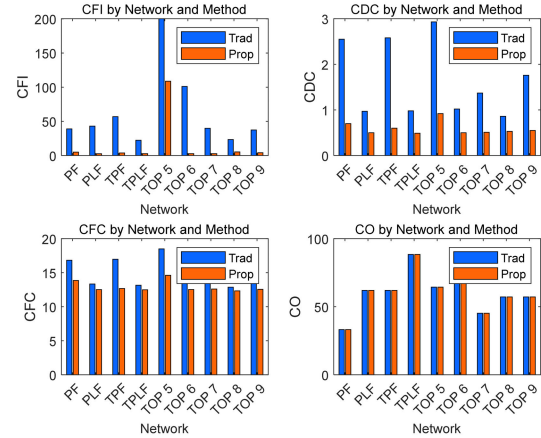


Fig. 12. Simulation Results: Prop vs. [35] Control during Random Intermittent Disconnections.

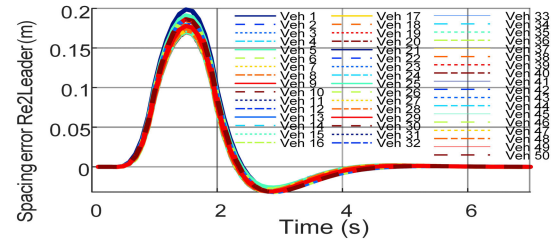


Fig. 13. Spacing error for 50 vehicles.

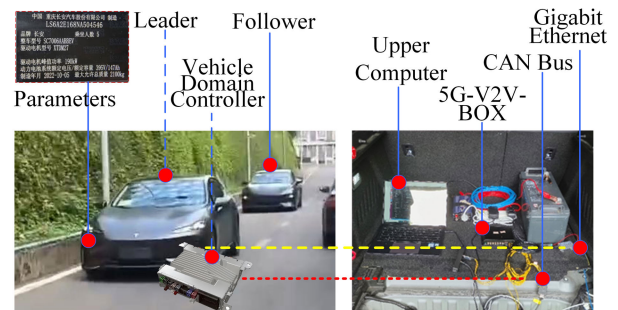


Fig. 14. Real-world experimental scenario.

4900 mm, as shown in Fig. 14. The 5G-V2V-BOX, featuring V2V communication capabilities, controls the vehicle by transmitting signals for torque, speed, acceleration, and gear position. The controller is equipped with an NXP i.MX6Q processor and a ZM8350 V2X module. The vehicle controller employs the NLopt optimization library, with a maximum optimization time of 15ms and an error tolerance less than $1e-4$. The 5G-V2V-BOX is illustrated in Fig. 15.

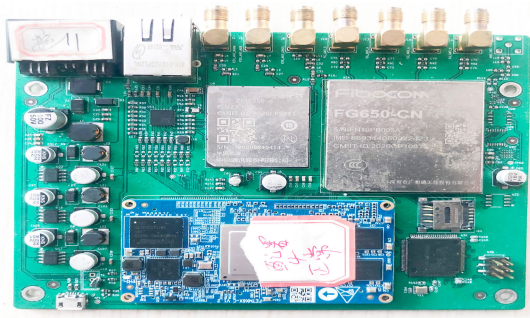


Fig. 15. Appearance of 5G-V2V-BOX controller.

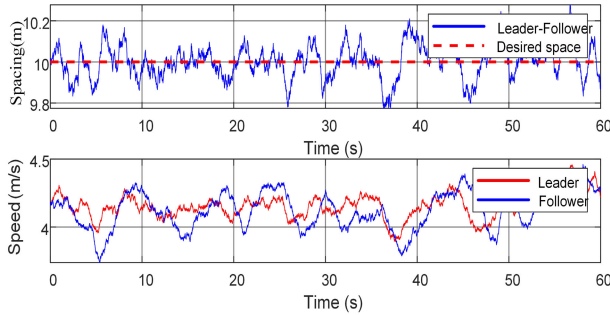


Fig. 16. Testing results: speed tracking and spacing maintenance.

Simulated network interruptions were introduced between 30 and 31s by implementing a counter that temporarily disabled the use of V2V messages, allowing an assessment of the spacing behavior under communication loss. The experimental results indicate that the follower vehicle successfully tracked the leader's speed of 15km/h (4.29m/s) while maintaining the desired spacing of 10m, as shown in Fig. 16. The maximum spacing error remained within 0.3 m despite actuator constraints (vehicle speed ≤ 30 km/h, laccelerationl $\leq 0.7\text{m/s}^2$), while the CDC was 0.26 and the equivalent CFC was 0.0144L.

VII. CONCLUSION

This study tackles the challenge of maintaining effective tracking in vehicle platoons under fluctuating V2V networks and random interruptions. We propose a platoon control method based on V2V network performance and the collaboration status estimation to adapt to dynamic network conditions. Platoon performance is assessed across four criteria: tracking, comfort, energy economy, and communication overhead. Three case studies are conducted, focusing on 15 vehicles, nine network topologies, fixed-time and random network interruptions. For the PF topology, the proposed method improves tracking, comfort, and energy economy by 87.1%, 72.5%, and 23.2%, respectively, compared to a baseline control methods. Results confirm the method's effectiveness in fluctuating network scenarios.

Although we performed extensive simulations and preliminary real-world experiments, this study did not consider the impact of dynamic topology switching on the controller. Additionally, the experimental vehicles are controlled by multi-level controllers rather than relying solely on the 5G-V2V-BOX in

real-world tests. Future work will address topology switching and conduct more comprehensive experiments to further improve the platooning control method.

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